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A An additional comparison with related works

Our focus in Section 2 is on an empirical illustration of the implicit regularization phenomenon, hence we refrain from attempting to prove this general conjecture, which we believe is a challenging task. The existing theoretical works [Blanc et al. \(2020\)](#); [Li et al. \(2021b\)](#); [Damian et al. \(2021\)](#) present two major weaknesses; they are essentially limiting analysis and as such fail at capturing the *entire optimization trajectory* and they primarily target regression tasks. The powerful mathematical framework for scale-invariant networks developed by [Li & Arora \(2019\)](#); [Li et al. \(2020\)](#) allows them to study in detail the benefits of normalization and its interplay with weight decay. By means of this framework, they state a fast equilibrium conjecture, which gives qualitative guarantees for the speed of convergence of the stochastic process to the stationary distribution in function space. They disentangle the evolution of the norm and the direction of the parameters and show how the evolution of the direction only depends on the intrinsic LR $\lambda_i = \eta\lambda$. However, a qualitative description of the stationary distribution, its dependence on this intrinsic LR and the relationship with generalization is missing [Li et al. \(2020\)](#). We attempt to fill this gap by providing a qualitative depiction of the stationary distribution and its dependence on the intrinsic LR shading some light towards understanding the relationship with generalization. The work of [Kodryan et al. \(2022\)](#) reports a similar observation, where the best test loss is achieved at a LR where the loss neither converges nor diverges but does not provide any explanation.

Table 1: Comparison of our work with closely related works on regression and implicit regularization phenomenon induced by noise in the algorithm.

Paper	Loss function	Algorithm	Implicit regularization
Damian et al. (2021) & Li et al. (2021b)	Squared loss & CE + label smoothing	Label noise GD	Trace of Hessian
Blanc et al. (2020)	Squared loss	Label noise GD	Jacobian norm
Li et al. (2020)	Scale-invariant loss	SGD	-
Andriushchenko et al. (2023)	Squared loss	SGD with large LR	Jacobian norm
Our work	Regularized CE	SGD with large LR	Jacobian norm

B Training details

Full experimental details are available in our public repository <https://github.com/tml-epfl/why-weight-decay> but we also list the main training details here. All the experiments are conducted for 3 different random seeds, the error-bars report one standard deviation. **CIFAR-10/100 experiments.** We train a VGG network without BatchNorm and preactivation ResNet-18 on CIFAR-10 and ResNet-34 on CIFAR-100 without data augmentations. We use standard SGD *without momentum* for all experiments. We note that ℓ_2 regularization and weight decay are exactly the same in this case. We use the standard He initialization ([He et al., 2015](#)) for all parameters. To make ResNets scale-invariant, we follow the approach of [Li et al. \(2020\)](#) consisting of fixing the last layer, removing the learnable parameters of the normalization layers and adding a normalization layer in the skip connection. For the experiments in Fig. 11, VGG is trained with LR = 0.1 and LR = 0.01 and weight decay parameter is fixed to be either $\lambda = 0.0$ or $\lambda = 0.008$. The ResNet-18 is trained with LR = 0.08 and LR = 0.001 and $\lambda = 0.0$ or $\lambda = 0.0125$. The ResNet-34 is trained with LR = 0.15 and LR = 0.001 and weight decay parameter $\lambda = 0.0$ or $\lambda = 0.01$. The total number of epochs is 1000 in all experiments in Fig. 11 and all the LR are decayed at epoch 500 to 0.0001. For the experiments in Fig. 15 we use scale-invariant ResNet-18 and project the SGD iterates on the unitary sphere. We test the following LR in the large-LR phase (0.0001, 0.0005, 0.00075, 0.001, 0.002, 0.003, 0.004, 0.005) to show different generalization performance. After 100 epochs all the learning rates are decayed to the same value 0.0001. In Fig. 15 we fine-tune every 2 epochs for 100 additional epochs with LR=0.0001. To measure the Norm of the Jacobian or the Trace of the Hessian we use a subset of 5000 training datapoints. Each run requires approximately 2 GPU hours on an Nvidia A100 GPU.

Tiny-ImageNet experiments. We train Resnet-18 without data augmentation. We use standard SGD *without momentum* in all our experiments. We use the following learning rates (0.0005, 0.0010, 0.0050, 0.0100, 0.0500, 0.1000, 0.1500, 0.2000, 0.2500) and weight decay parameter (0.0200, 0.0150, 0.0125, 0.0100, 0.0075, 0.0050, 0.0025, 0.0010, 0.0005, 0.0000). To measure